

# ZOIBA TABASSUM

Full-Stack Development | AI / ML Engineering | BE Software Engineering · NUST · 6th Semester

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## SUMMARY

6th-semester Software Engineering student at NUST with production-level full-stack experience (JavaScript, Node.js, React, PostgreSQL) and applied ML/DL knowledge from university coursework and independent projects. Built real-world data systems at WWF internship. Developing a multi-Agent AI platform for SME digitization as Final Year Project. 100 Days of LeetCode 2025 badge.

## TECHNICAL SKILLS

|           |                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------|
| Languages | Python · JavaScript (ES6+) · C++                                                                                    |
| AI / ML   | scikit-learn · PyTorch · NumPy · Pandas · Multi-Agent Systems · LLM APIs · LangChain · Data Analysis (foundational) |
| Web & API | React.js · Node.js · Express.js · REST APIs · HTML5 · CSS3 · Tailwind CSS                                           |
| Databases | PostgreSQL · MySQL · Relational Schema Design                                                                       |
| Tools     | Git · GitHub · Colab Notebook · Vite · GSAP · Framer Motion                                                         |

## PROJECTS

### AI-Driven SME Digitization Platform [Final Year Project — In Progress] · Python · LangChain · Crew AI · Fast API · PostgreSQL · React

- Multi-Agent AI system that autonomously discovers Pakistani SMEs, scores digital readiness via an ML classifier across 5 dimensions and generates personalized roadmaps using a RAG pipeline over a ChromaDB vector store.
- Agentic pipeline: Discovery → Readiness Scoring → Recommendation (RAG) → Prototype Generation → Personalized Outreach — zero human intervention after initialization.

### AIRYZE — Intelligent AQI Monitoring System · React · Node.js · Express · PostgreSQL · External APIs

- Real-time AQI data ingestion pipeline storing time-series pollutant data in PostgreSQL; automated Gmail health-alert system triggered by user-defined thresholds; interactive map and trend visualizations.

### ML & DL Labs — University Coursework Repo · Python · scikit-learn · PyTorch · NumPy · Pandas · Colab

- Collection of hands-on labs and mini-projects from NUST ML and Deep Learning electives — covering supervised learning, neural networks, model evaluation, and data preprocessing using Python, scikit-learn, and PyTorch.
- Implemented and compared classification and regression algorithms (linear/logistic regression, decision trees, SVMs, k-NN) with NumPy/Pandas pipelines and Matplotlib visualizations on real datasets.

### Farm Data Management System [WWF Internship] · Node.js · Express · MySQL · HTML · CSS

- Role-based full-stack platform managing farmers, crops, livestock, and land records — replacing paper workflows; normalized MySQL schema enabling real-time admin reporting.

### Portfolio Website · React.js · Tailwind CSS · Vite · GSAP · Framer Motion

- Production-grade animated portfolio with scroll-triggered GSAP interactions and Framer Motion transitions — demonstrates front-end engineering depth beyond functional correctness.

## EXPERIENCE

### World Wide Fund for Nature (WWF) — Technical Intern

Jul – Aug 2025

Food & Agriculture Sector · Islamabad

- Designed and deployed a full-stack agricultural data system (Node.js, MySQL) replacing manual workflows; engineered a normalized schema for structured multi-variable data collection and admin reporting.
- Collected, cleaned, and analyzed agricultural datasets to support sustainable farming decisions — an applied data analysis workflow directly informing program reporting.

## EDUCATION

### National University of Sciences and Technology (NUST)

2023 – 2027

BE Software Engineering · Islamabad · Relevant: Machine Learning, Deep Learning, DSA, Database Systems

## CERTIFICATIONS & ACHIEVEMENTS

100 Days of LeetCode 2025 (Badge) · The Complete JavaScript Course 2025 — Jonas Schmedtmann, Udemy (In Progress) · WOH Virtual Youth Leadership Program 2025 · Volunteer, NUST Open House 2025 & 2026